

Algebraic Methods for Matrix Quantum Mechanics with Orthogonal Polynomials

Abstract:

When using matrix methods in quantum mechanics it is common to use orthogonal polynomials as an expansion basis. The eigenvalues and eigenvectors of such provide the energies and wavefunctions for bound states in the Schrodinger equation, and when considering a complex scaling method we can extract the lifetimes of resonance states. Investigating the properties of orthogonal polynomials can provide insights into more computationally efficient methods for constructing the necessary matrices. We present results on the application of these methods to a number of simple test cases of 1D problems where we can compare to standard theory.

Introduction -

Matrix methods of solving the Schrodinger equation involves solving the Schrodinger equation in the following form,

$$\sum_{n=0}^{\infty} \langle m|H|n\rangle \langle n|\psi_i\rangle = E_i \langle m|\psi_i\rangle,$$

this is an eigenvalue eigenvector equation for a matrix. Computationally it is impossible to do infinite sums so approximate by truncating the matrix to be $N \times N$. We are left to calculate these sums which in position space this involves integrals of the form,

$$\langle m|H|n\rangle = \int_a^b \langle m|x\rangle H \langle x|n\rangle dx,$$

where the $\langle x|n\rangle$ are the orthogonal polynomials multiplied by normalization and weight function called, $\Phi_n = N_n f(x) P_n(x)$ and the Hamiltonian is given by choice of potential. Computationally we would like to avoid doing these integrals, such as using the ladder operators but now more generally with any orthogonal polynomial. Choice of polynomial fixes the boundaries,

Polynomial	a	b
Hermite	$-\infty$	∞
Associated Laguerre	0	∞
Gegenbauer	-1	1

Method -

When considering Hamiltonians, we often want to solve we are given them in terms of kinetic and potential terms. The kinetic term involves derivatives. The potential term can take any kind of appropriate function of x , depending on the problem. Therefore, the task becomes to find methods of solving integrals involving derivatives, and any functions of x algebraically to which we can give back to a computer to construct Hamiltonian matrices.

Firstly, for the most common potentials the power series, we are given a powerful recursion relation of all the orthogonal polynomials to solve this,

$$x\Phi_n(x) = a_n\Phi_{n+1}(x) + b_n\Phi_n(x) + c_n\Phi_{n-1}(x),$$

Expanding the method to any function of x in general, we can use a method of diagonalisation to cast the function of x onto the eigenvalues only,

$$V(A) = TV(D)T^{-1},$$

Using the diagonalisation technique then becomes taking the eigenvectors between the function we are interested in acting on the eigenvalues, and thus we have a method of finding any potential terms without using integration.

When considering the kinetic term, there exists derivatives acting on both the orthogonal polynomial and the weight function, which is necessarily a function of x ,

$$\frac{d}{dx}\Phi_n(x) = N_n f'(x)P_n(x) + N_n f(x)P'_n(x),$$

there exists similar universally common recursion relations for the derivatives of orthogonal polynomials in terms of the orthogonal polynomial. However, there is some work needed to deduce an approach to the $f'(x)$, but we can use some tricks and eliminate this term, finding the complete matrix found using the anti-symmetry.

The second derivative is found in terms of the first derivative and can be found recursively.

Proof that this method works can be found in examining simple test cases with known results. We use the variational principle to show that the method developed converges on the expectation answer in the case of some test Hamiltonian.

Future Work -

For Hamiltonians which generate resonance the method of complex scaling both the $\hat{p}$ and $\hat{x}$ operators with,

$$\begin{aligned}\hat{p} &\rightarrow \hat{p}e^{-i\theta}, \\ \hat{x} &\rightarrow \hat{x}e^{i\theta},\end{aligned}$$

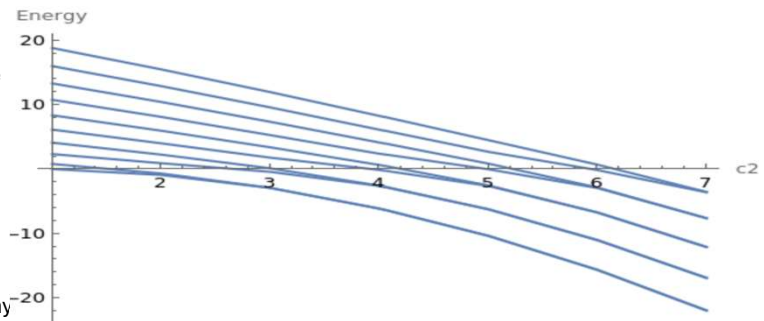
has the effect of transforming our resonance problem into a standard "bound-like" eigenvalue problem i.e. the resonance states become normalisable, allowing us to use standard techniques to solve the problem such as solving the Schrodinger Equation. Finding that the eigenvalue calculated are given by,

$$E = E_0 - i\frac{\Gamma}{2}.$$

Test Cases-

Double Well Potential: $H = \frac{\beta^2}{2}P^2 - \frac{c_2}{2}X^2 + \frac{1}{\beta^4}X^4$

The theory here tells us the eigenvalues below a certain level determined by the depth size (here controlled with c_2) will pair up as parity doublets then as usual split up above the well.



The depth size c_2 controls if the eigenvalues pair up.

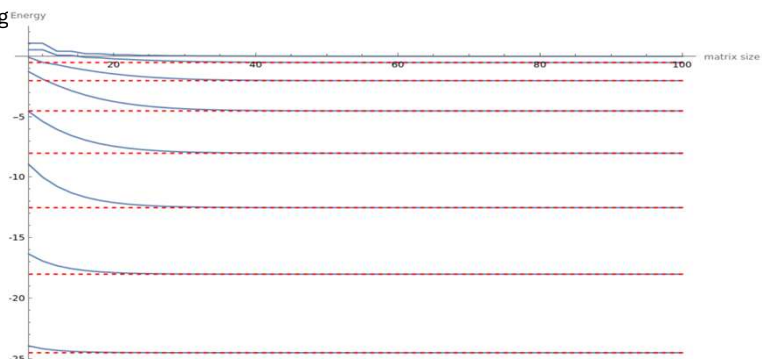
Pöschl-Teller Potential: $H = \frac{p^2}{2m} - \frac{\lambda(\lambda+1)}{2}(\text{sech } x)^2$

There exists both bound states and a continuum of states above the bound states here. The theory tells us that the bound states are found by,

$$E = -\frac{\mu^2}{2},$$

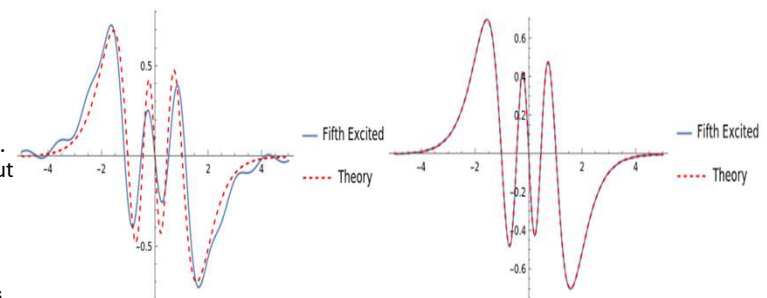
$$\lambda = 1, 2, 3, \dots,$$

$$\mu = 1, 2, \dots, \lambda - 1, \lambda.$$



The variational method here demonstrates that at the eigenvalues converge onto the theory with higher matrix size.

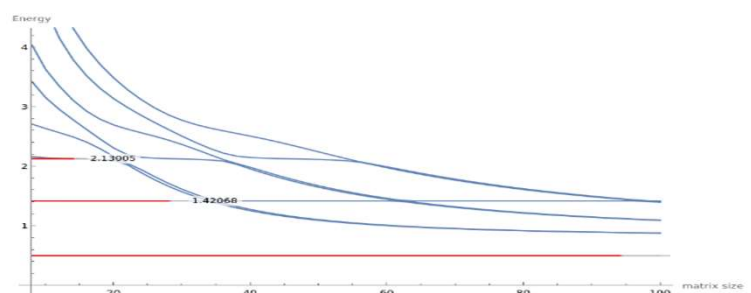
$\lambda = 7$ plotted against the theory.



The plot on the right uses a higher matrix size which is closer to theory.

Resonance Potential (1) $-H = \frac{1}{2}P^2 + (\frac{X^2}{2} - J)e^{-\lambda X^2}$

A simple potential in which resonances arise. When plotting for the variational principle here, visually it is possible to pick out the resonances.



References:

1 - H J Korsch and M Glück 2002 Eur. J. Phys. 23 413